

Vector-Bench: Repairing SVGs Without Breaking Them

Yug Aditi Gupta*
yug@thetalab.tech
Theta Labs

Prannay Hebbar*
prannay@warping.co
Warping

*Equal contribution; the paper and accompanying material are shared work.

Abstract

Instruction-based vector editing requires two capabilities: making a requested change and leaving everything else alone. The second is easy to miss when an output is judged only as a raster image. We introduce **Vector-Bench**, a compact, difficult benchmark of 40 SVG repair tasks. Each task pairs a corrupted SVG program with an author-written visual instruction, a hidden target program, 5.05 annotated repairs on average, and an average of 60.55 protected objects. Instructions describe visible defects without exposing element identifiers, coordinates, color codes, or path data. We define a deterministic binary specification reward: requested repairs use attribute-aware perceptual tolerances, while unrequested rendering- or application-relevant structure must remain semantically unchanged and the result must be a valid SVG. Canonical target equality and stricter source fidelity are retained as diagnostics. Validity-gated repair progress, a near-complete tier, and valid-output Unintended Change Rate (UCR) explain partial outcomes. We evaluate 34 model endpoints (25 listed as open-weight, 5 inexpensive controls, and 4 frontier closed endpoints) over 1360 requests. The strongest endpoint reaches only 15.0% full specification success, despite 43.7% mean repair progress, showing that apparent repair progress and specification-faithful editing remain substantially different. All prompts, outputs, scoring code, costs, and per-task reports are released.

1 Introduction

An instruction such as “the amber signal has gone dark; fix it and leave the station alone” defines both a positive requirement and a large set of negative requirements. The signal must change, while the train, clock, cables, platform, and every unrelated program detail must not. A model can produce a plausible raster while renaming elements, rewriting

path data, shifting an unrelated object, or deleting metadata. These are not cosmetic differences for an SVG: they alter an executable, structured artifact ([World Wide Web Consortium, 2018](#)).

Most generation metrics emphasize final appearance. That is appropriate for open-ended synthesis, but insufficient for local editing. A repair benchmark must answer three separate questions: Did the requested objects change? Did unmentioned objects remain stable? Is the resulting program valid and usable? Vector-Bench makes all three observable through hidden target SVGs and explicit protected-object sets.

The benchmark is deliberately small and dense rather than broad and shallow. Its 40 tasks contain 202 annotated repairs across station, harbor, campsite, market, and larger scenic illustrations. The public instruction never names a DOM identifier or serialized numeric/path value. Solvers must interpret the visible scene and edit the source accordingly. Figure 1 shows representative inputs and targets.

Our contributions are:

1. a frozen corpus of dense, author-described SVG repairs with explicit intended edits and broad preservation surfaces;
2. a transparent executable evaluator whose binary specification reward combines tolerant requested-edit checks, rendering- and application-relevant semantic preservation, and SVG validity, while retaining source fidelity separately;
3. a reproducible 34-endpoint evaluation with response provenance, token usage, cost, failures, and model-produced SVGs; and
4. an empirical account of the gap between completing visible repairs and preserving the full vector program.

Corrupted input

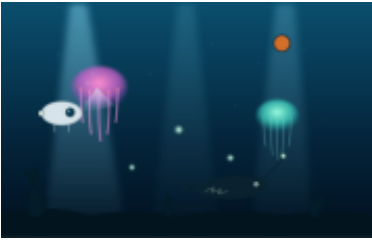


Hidden target



The amber signal has gone dark, the train door is sitting too far to the right, and the overhead cable has an awkward sag. Put those three details back in place and leave the rest of the station alone.

Corrupted input

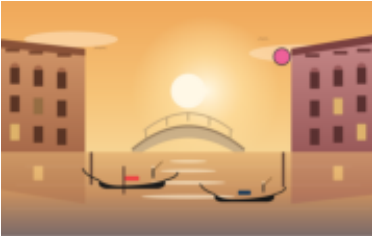


Hidden target



In the deep-sea scene, remove the stray colored marker and move the displaced glowing particle back toward the nearby cluster. The large jellyfish should have fine tentacles and a soft pink lower rim, while the second shaft of light should be clearly visible. Leave the other creatures alone.

Corrupted input



Hidden target



Remove the colored marker from the Venetian sunset. Center the large sun glow behind the canal, restore the dark red body of the gondola, slim the main arch of the bridge, and bring back the warm glow in the affected window. Keep the other buildings and reflections intact.

Figure 1: Representative corrupted inputs and hidden targets. Public prompts describe visible mistakes in ordinary language without naming source IDs or revealing target numeric/path values.

2 Related Work

Vector generation. IconShop generates vector icons from text (Wu et al., 2023); StarVector generates SVG code from images and text (Rodriguez et al., 2023); and LLM4SVG studies complex vector understanding and generation with language models (Xing et al., 2024). These systems establish SVG as a useful meeting point between visual generation and program synthesis. Vector-Bench focuses on local repair rather than open-ended generation.

Instruction-based SVG editing. SVGEEditBench introduced quantitative evaluation for LLM-based SVG editing (Nishina and Matsui, 2024), and SVGEEditBench V2 expanded instruction-based editing evaluation (Nishina and Matsui, 2025). VectorEdits provides a larger dataset and benchmark for vector-graphics editing (Kuchař et al., 2025). Our scope is complementary: we emphasize dense multi-repair scenes, tolerance-aware repair checks, semantic preservation outside the requested fields, and publication of every model output. The hid-

Statistic	Value
Tasks	40
Annotated repairs	202
Repairs per task (mean)	5.05
Protected objects per task (mean)	60.55
SVG elements per task (mean)	85.4
Input characters (total)	294,798

Table 1: Frozen corpus statistics. Element counts include non-rendering SVG nodes; protected-object counts refer to identified scene objects.

den target anchors deterministic checks without requiring a model to reproduce every requested coordinate or color exactly.

3 Benchmark

3.1 Task Format

Each task is a tuple

$$\tau = (x, S_0, S^*, E, U),$$

where x is an author-written repair instruction, S_0 is the corrupted SVG, S^* is the hidden target, E is a set of expected object-attribute repairs, and U is the scene-object subset used for object-level preservation diagnostics. The whole-document preservation gate covers every unrequested node, including nodes outside U . The model receives only (x, S_0) . Target source, expected diffs, target-part lists, and preserve lists are excluded from the prompt and recorded as hidden evaluator inputs.

Instructions were rewritten after task freezing by the authors. They describe visible objects and relations (for example, a moon that drifted down and left or a jellyfish whose tentacles became too heavy) without SVG IDs, numeric coordinates, hex colors, path commands, or attribute names. A corpus regression hash covers every non-instruction task field, ensuring this language-only rewrite did not change inputs, targets, annotations, or ordering.

3.2 Composition and Difficulty

The corpus contains 20 compact but busy scenes derived from four authored scene templates and 20 larger scenic illustrations. Ten tasks are marked hard and 30 very hard. Tasks require between three and nine repairs, including insertion or deletion, recoloring, displacement, path restoration, line-weight correction, and opacity restoration. Every unrequested node is covered by the semantic preservation gate; U enumerates the identified scene objects used for UCR.

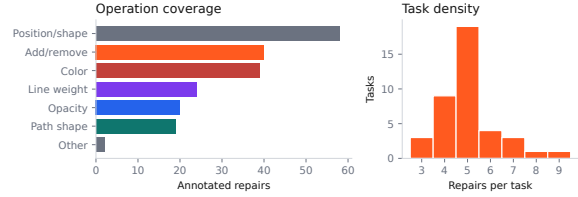


Figure 2: Repair operations and per-task edit density.

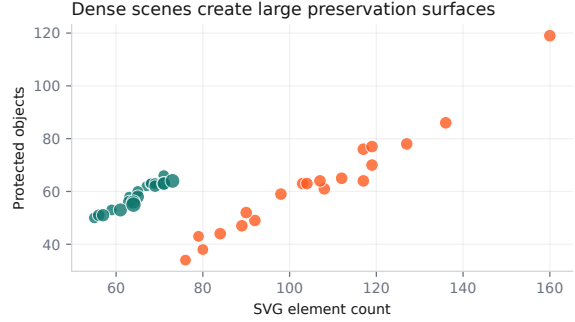


Figure 3: SVG size and preservation surface. Marker area scales with the number of requested repairs.

4 Executable Evaluation

4.1 Three-Gate Specification Reward

Let $C(S)$ parse an SVG and canonicalize representation-only variation. We discard namespace spelling differences and attribute order, normalize numeric spellings and separators in paths, points, transforms, and view boxes, and map common equivalent color spellings (for example, white, #fff, and $\text{rgb}(255, 255, 255)$) to one form. A second form $V(S)$ represents rendering- and application-relevant semantics: consistently renamed IDs and their local references are aligned by tree position, and equivalent inline-style and presentation-attribute storage is merged. Element order, nesting, geometry, paint, text, and unresolved references remain strict. Malformed XML has neither form. Requested objects are resolved by ID first and by stable tree position when an otherwise equivalent editor has renamed IDs.

For each requested edit $e \in E$, an attribute-aware predicate $\text{ok}_e(S)$ compares the produced value with the hidden target. Existence and non-visual values are exact. Scalar geometry uses absolute error; colors use CIE Lab ΔE_{76} ; and point lists use coordinate RMS. Paths are sampled at 129 length-normalized positions and compared with a symmetric nearest-point distance combining RMS and the 95th percentile. This admits the same visible curve expressed with different SVG commands. For matching command streams, coordinate RMS remains an additional acceptance route. When the

corrupted-to-target distance is δ_e , the tolerance is bounded by

$$t_e = \min(c_e, \max(f_e, r_e \delta_e), 0.9 \delta_e),$$

so the known corrupted value cannot pass. Scalar checks use $r_e = 0.50$, color uses 0.45, point lists use 0.55, and paths use 0.70. Position caps are 2.5% of the relevant viewport dimension, color is capped at $18 \Delta E_{76}$, opacity at 0.12, and path/point geometry at 3% of the smaller viewport dimension. Stroke width is capped at one user unit or 40% of target width, whichever is larger. Floors avoid rejecting immaterial numeric noise. Every per-check report exposes target distance, tolerance, baseline corruption distance, and progress.

Approximation is permitted only at requested fields. Let M_E mask those fields in both the produced and target trees (and remove requested insertion/deletion sites). The primary preservation gate requires $V(M_E(S)) = V(M_E(S^*))$. It accepts consistent ID renaming and style-storage rewrites but catches changes to unlisted definitions, anonymous nodes, root attributes, element order, and unrequested visual attributes. We separately report the stricter source predicate $C(M_E(S)) = C(M_E(S^*))$. The validity gate requires a complete SVG root, well-formed XML, unique nonempty IDs, valid path, point-list, transform, and view-box syntax, and resolvable local URL and href references.

The primary reward remains binary:

$$\begin{aligned} R_{\text{spec}}(S) = & \mathbb{K}[\text{valid}(S)] \\ & \cdot \mathbb{K}[\forall e \in E : \text{ok}_e(S)] \\ & \cdot \mathbb{K}[V(M_E(S)) = V(M_E(S^*))]. \end{aligned}$$

Canonical target equality $\mathbb{K}[C(S) = C(S^*)]$ and source preservation are diagnostics and are not required for a pass. No learned judge or model-based grader appears in the public protocol.

4.2 Evaluator Audit

The evaluator is tested as an executable artifact rather than treated as an unexamined source of labels. The target SVG must receive reward one; the corrupted input must fail at least one requested repair; and unintended deletions, unrequested attribute changes, duplicate identifiers, broken references, malformed paths, and unknown added elements must fail the preservation or validity gate. Regression tests also cover approximate numeric and color repairs, alternate path command topologies, equivalent style storage, and consistent identi-

fier renaming. The frozen corpus has a regression hash over every non-instruction task field. These checks establish implementation invariants, not human perceptual validity: the comparator’s scope and the absence of a human agreement study remain limitations.

Table 2 reports four deterministic controls over all tasks. Returning the corrupted input remains valid and perfectly preserved but earns no repair credit. Copying the hidden target is the evaluator upper bound. Adding one protected metadata mutation to each target preserves all requested repairs but forces the binary reward to zero, and malformed targets receive neither validity nor progress credit. These are evaluator controls, not solver baselines: the target is unavailable to evaluated models.

4.3 Diagnostic Metrics

For expected checks E , tolerance-aware edit completion is

$$\text{Edit}(S) = \frac{1}{|E|} \sum_{e \in E} \mathbb{K}[\text{ok}_e(S)].$$

For valid outputs and checks with a distance, repair progress is $q_e = \text{clip}(1 - d_e/\delta_e, 0, 1)$; exact and existence checks use binary progress. Invalid outputs set every q_e to zero, preventing a missing document from receiving deletion credit. We report $|E|^{-1} \sum_e q_e$ in addition to completion. A *near-complete* output is valid and semantically preserved, misses at most one requested check, and has at least 80% mean progress. It still receives zero binary reward.

For protected objects U , preservation and unintended-change rate are

$$\text{Pres}(S) = \frac{1}{|U|} \sum_{u \in U} \mathbb{K}[V(u_S) = V(u_{S^*})],$$

$$\text{UCR}(S) = 1 - \text{Pres}(S).$$

We additionally report all-repairs pass rate, whole-document semantic-clean rate, source-preservation rate, SVG validity, whitespace-normalized source-string equality to the target, canonical target match, provider errors, scheduled elapsed time, tokens, and cost. Model-level UCR is averaged over valid outputs so truncation is not mislabeled as collateral editing; it is undefined when a model has no valid output. A provider error or invalid output receives reward and repair progress zero.

Control	Full	Repair	Progress	Clean	UCR	Valid
Return corrupted input	0.0	0.0	0.0	100.0	0.0	100.0
Hidden target copy	100.0	100.0	100.0	100.0	0.0	100.0
Target + protected mutation	0.0	100.0	100.0	0.0	1.7	100.0
Malformed target	0.0	0.0	0.0	0.0	–	0.0

Table 2: Deterministic evaluator controls over all 40 tasks. Values are percentages; UCR is conditional on valid SVGs and is undefined when none are valid. The protected-mutation control verifies that completing every repair cannot compensate for an unrequested change.

Model	Full ↑	Near ↑	Repair ↑	Progress ↑	Clean ↑	UCR ↓	Valid ↑	Cost
Claude Sonnet 5	15.0	2.5	15.0	43.7	42.5	0.8	62.5	\$2.032
KAT Coder Air V2.5	12.5	0.0	15.0	27.3	25.0	0.3	42.5	\$0.128
MiniMax M3	10.0	0.0	10.0	33.0	25.0	1.0	45.0	\$0.240
HY 3	7.5	2.5	10.0	56.8	37.5	1.8	100.0	\$0.092
Gemma 4 31B IT	5.0	5.0	10.0	54.8	47.5	0.8	87.5	\$0.084
Qwen3.5 397B A17B	5.0	0.0	5.0	39.8	32.5	1.4	65.0	\$0.618
DeepSeek V4 Pro	5.0	2.5	5.0	30.6	20.0	1.2	47.5	\$0.508
Kimi K2.6	5.0	5.0	5.0	16.0	15.0	0.3	25.0	\$1.077
Qwen3 Coder	2.5	0.0	5.0	54.7	45.0	1.4	100.0	\$0.295
Gemini 3.1 Flash Lite	2.5	0.0	7.5	54.5	25.0	2.5	100.0	\$0.269

Table 3: Top ten endpoints under binary specification reward. Full, Near, Repair, Progress, Clean, UCR, and Valid are percentages; UCR is conditional on valid outputs, and cost is provider-reported for all 40 tasks. Full results are in Appendix B.

5 Model Study

5.1 Protocol

We evaluate 25 endpoints designated as open-weight in the study manifest, 5 inexpensive closed controls, and 4 frontier closed endpoints through the OpenRouter chat-completions API (OpenRouter, 2026). These designations record the endpoint grouping used for analysis and are not an independent license audit. The exact endpoint IDs appear in Appendix B.

Every model receives the same system instruction, author-written repair request, and corrupted SVG source. We record one scored outcome per task and omit temperature rather than imposing a provider-incompatible value. In this run, input-adaptive output ceilings range from 4,096 to 7,002 tokens. The runner retries rate limits and transient server failures against the same endpoint. It never falls back to another model; the client permits two transport retries inside each harness attempt. We record requested and resolved model IDs, response ID, finish reason, prompt, completion and reasoning tokens when supplied, scheduled wall-clock elapsed time, and provider-reported cost. Elapsed time begins when a model–task job is scheduled and therefore includes local concurrency-queue wait; we expose it for run auditing, not as endpoint latency. Each source cohort uses a reservation-based budget guard capped at \$25; the completed combined study cost \$14.88. The current runner ad-

ditionally emits credential-redacted, append-only request, retry, response, extraction, and scoring events. Because the reported study predates this complete attempt logger, its public rows are marked `legacy_final_record`: the extracted final SVG (or malformed final body) and metadata are retained, but discarded retry envelopes and response wrappers cannot be reconstructed.

This is an endpoint comparison, not a general ranking of model intelligence. It uses one decoded response per model–task pair and does not include a human performance baseline. The released artifact supports exact reruns of the scored protocol, but does not estimate sampling variance, human performance, or performance on unseen SVG distributions.

5.2 Uncertainty

We report two-sided 95% Wilson score intervals over the 40 binary task outcomes for each model. Unlike a nonparametric bootstrap, Wilson intervals remain informative when a model records zero passes. They quantify uncertainty under a Bernoulli task-sampling interpretation only; they do not estimate decoding variance because each model–task pair has one scored outcome. Endpoint tables are ordered by full specification pass, mean repair progress, lower valid-output UCR, validity, and model name.

6 Results

Specification success remains rare. Only 32 of 1360 outcomes satisfy all three gates (2.35%), while 14 (1.03%) are near-complete and 679 (49.9%) make at least one annotated repair. Validity-gated repair progress is 27.5% over all scheduled outcomes and 50.9% conditional on valid outputs. Valid-output UCR is 2.4% overall. Across all outcomes, 3.24% satisfy every requested repair and 19.85% preserve the masked document semantically; the stricter source gate passes 19.78%, and only 0.07% canonically match the full hidden target. The strongest endpoint, Claude Sonnet 5, obtains 15.0% specification success. Its repair progress is substantially higher at 43.7%, while its valid-output UCR is 0.8% and SVG validity is 62.5%. The difference is the central benchmark result: models frequently perform some requested visual work but fail the full repair-and-preserve contract. Figure 4 reports Wilson intervals for every endpoint.

Frontier endpoints remain subject to the same bottlenecks. Within the frontier cohort, Claude Sonnet 5 ranks highest with 15.0% specification success and 43.7% repair progress. Its SVG validity is 62.5% and its truncation rate is 40.0%. Length-limited outcomes elsewhere in the cohort remain failures because the protocol requires a complete usable artifact.

The failure mass is mostly incomplete or unusable output. Figure 5 partitions every model-task pair into exactly one gate outcome. This decomposition prevents low binary success from being interpreted as one homogeneous failure mode: a valid partial repair, an all-repairs output with collateral changes, and an invalid or missing SVG all receive zero reward for different reasons.

The conclusion is stable to moderate tolerance changes. Table 4 rescales every distance-based tolerance while retaining exact structural checks, validity, semantic preservation, and the rule that the known corruption cannot pass. The absolute pass rate changes, as expected, but remains low under both stricter and looser settings. Thus the result is not created by requiring canonical target equality, although the precise binary rate remains calibration-dependent.

Repair and preservation are separate axes. Figure 6 (left) plots mean repair progress against

Scale	Full n	Full %	Repair n	Repair %
0.5×	11	0.81	13	0.96
1.0×	32	2.35	44	3.24
1.5×	46	3.38	67	4.93

Table 4: Tolerance sensitivity over the same 1360 recorded outcomes; the primary setting is bold. Rates are percentages. Scaling applies only to distance-based requested-edit tolerances; all other reward gates are unchanged.

valid-output UCR. A model can move right by correcting visible defects while also moving upward by rewriting protected scene objects. Reporting completion alone would hide this behavior; reporting only full success would hide useful partial capability.

Operation families expose different bottlenecks. Figure 7 decomposes the expected checks. Simple color or add/remove corrections need not predict path-shape or position restoration, especially because the prompt gives visual descriptions instead of literal DOM values. This decomposition also distinguishes an endpoint that attempted an edit but selected the wrong source element from one that returned an invalid or unchanged document.

Cost is not a sufficient capability proxy. The run spans inexpensive small endpoints and larger reasoning or coding models. Figure 6 (right) shows repair progress against total 40-task cost. It is a descriptive endpoint comparison rather than a hardware-efficiency claim: OpenRouter pricing and routing can change, and the study uses one scored outcome per task.

7 Error Analysis

Appendix C traces a full pass and Appendix D a full failure; both make the failure modes below concrete.

Literal partial repair. Models often identify an obvious semantic defect, such as a signal with the wrong color, while leaving a displaced object or malformed curve unresolved. These outputs earn partial edit-completion credit but specification reward zero.

Global rewrites. Some responses regenerate or reformat substantial portions of the SVG. The semantic normalizer accepts harmless formatting, equivalent style storage, and consistent ID renaming, but changed visible paths, reordered scene objects, broken references, or missing definitions

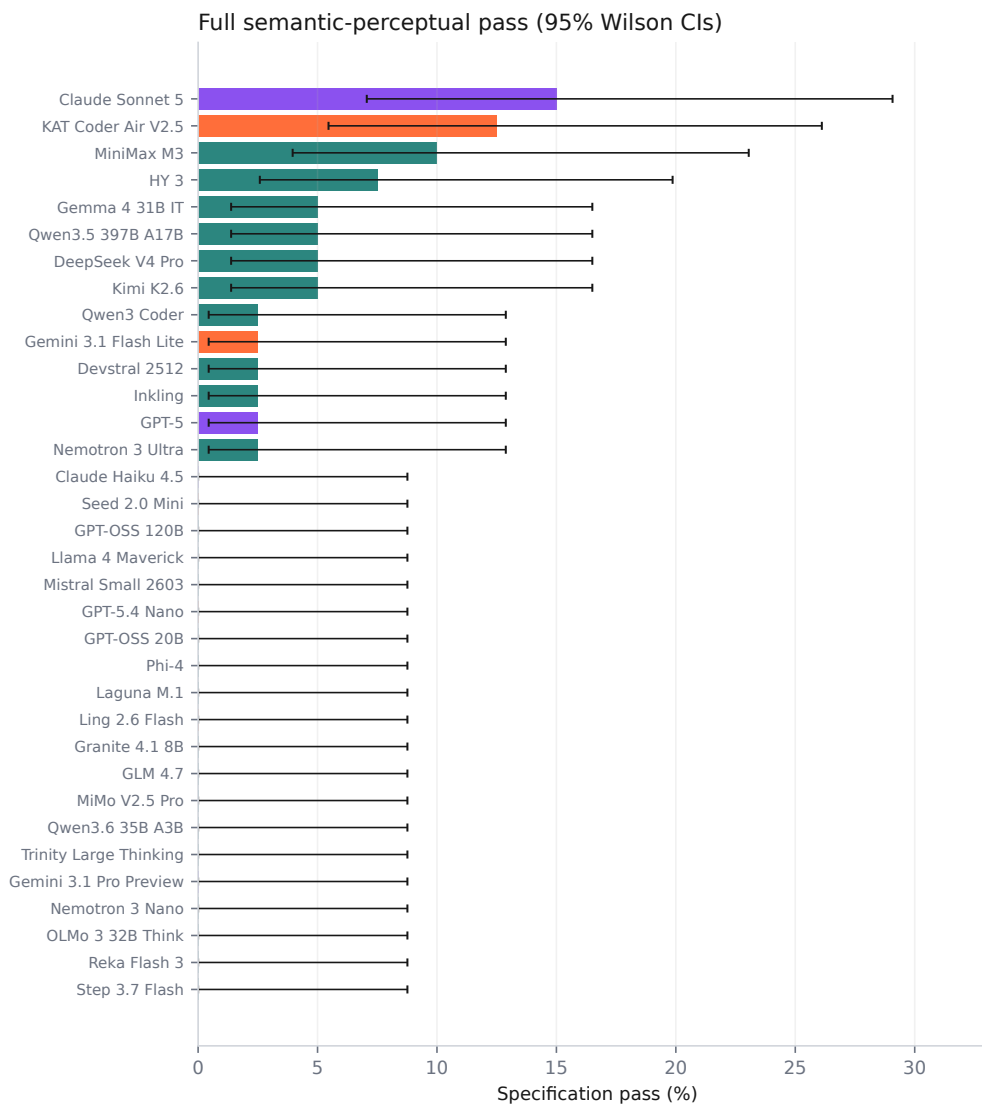


Figure 4: Binary specification success for all endpoints. Orange bars are inexpensive controls, teal bars are endpoints listed as open-weight, and purple bars are frontier closed endpoints.

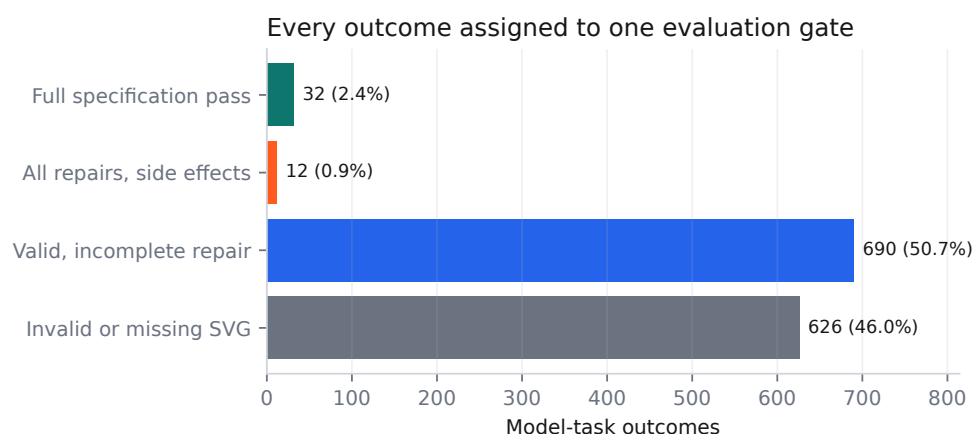


Figure 5: Mutually exclusive decomposition of all 1360 outcomes. The first two rows distinguish clean specification success from completed repairs with side effects; the remaining rows isolate incomplete valid edits from invalid or missing artifacts.

count as changes. The strict source-fidelity diagnostic still surfaces source rewrites when their visual semantics are preserved.

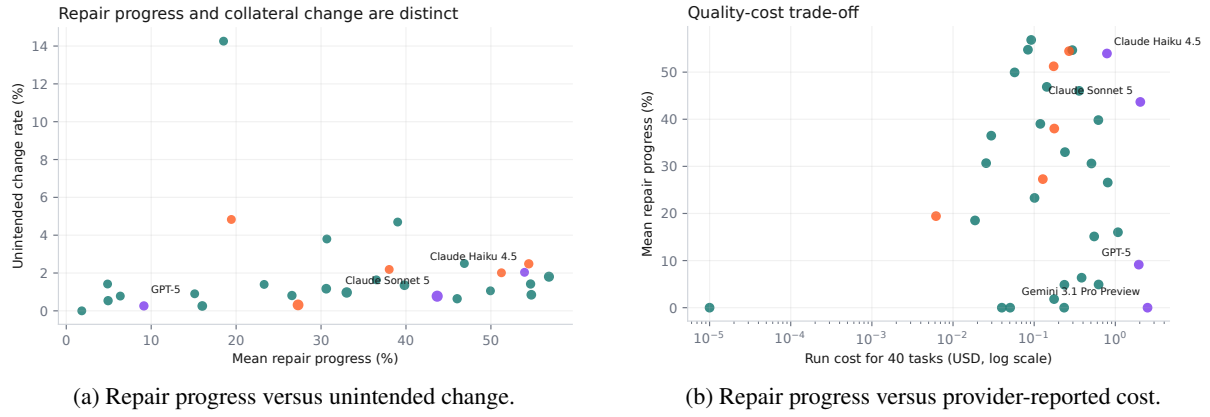


Figure 6: Complementary model-level diagnostics. In (a), marker area scales with specification pass rate and lower right is preferable. Cost in (b) is the recorded total for 40 tasks on a logarithmic axis.

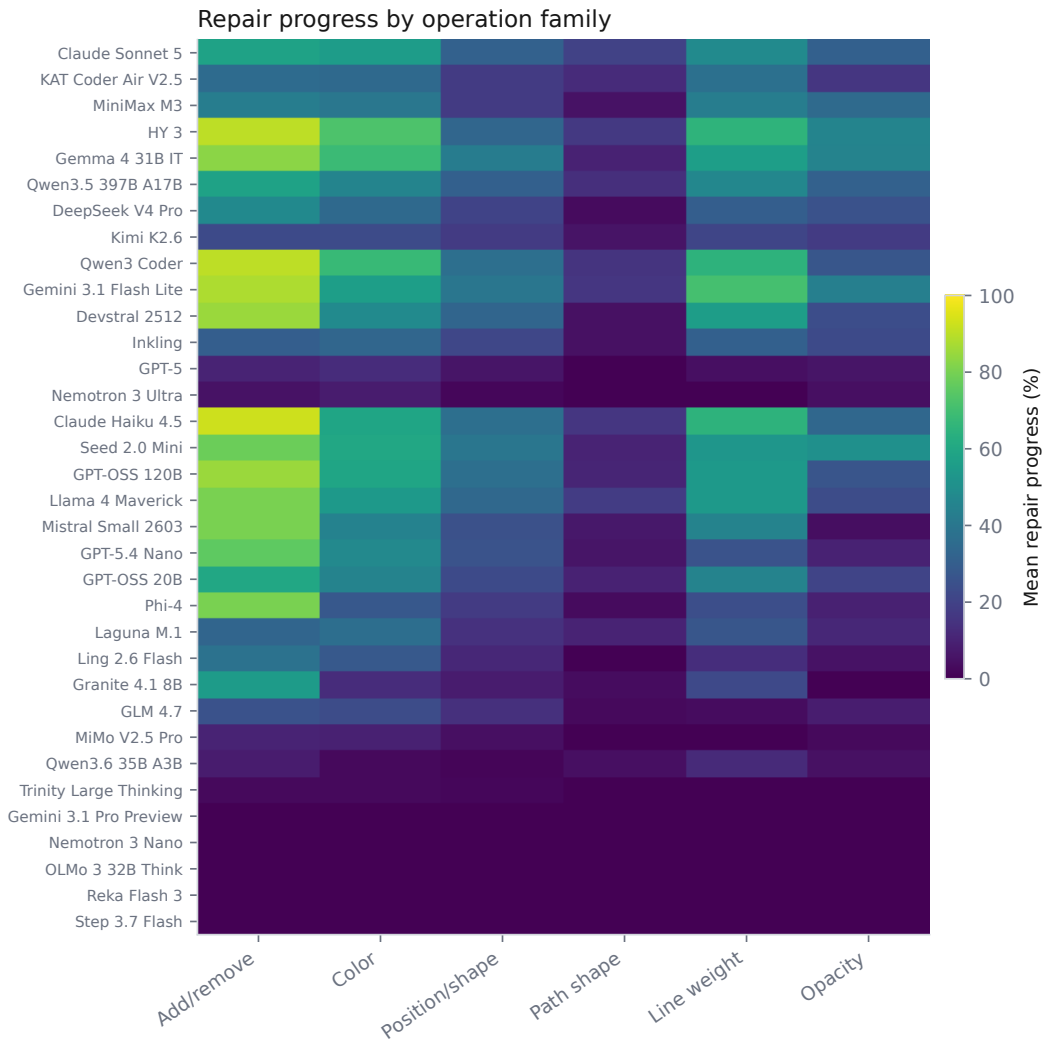


Figure 7: Mean expected-edit repair progress by operation family. Rows follow overall specification-pass rank.

Approximate repair without collateral change. Prompts intentionally omit exact coordinates, color codes, and path commands. The tolerance-aware repair gate accepts bounded alternatives to the hidden target, but only when the remainder of the SVG is unchanged. This separates a reasonable visual ap-

proximation from a global rewrite. The full-target match diagnostic quantifies canonical reproduction without controlling the primary reward.

Provider and truncation failures. Only 54.0% of scored outcomes pass structural SVG validity. The harness preserves raw malformed output, API

errors, and finish reasons in the released records. Transient failures are retried only against the requested endpoint; persistent failures remain zero-scored observations. The aggregate provider error rate is 3.0%, and 37.8% of all outcomes end at the provider’s output-length limit; parse failure is measured separately.

8 Limitations and Artifact Risks

Vector-Bench has only 40 tasks and should be treated as a focused stress test, not a comprehensive measure of visual editing. Public targets make future contamination possible. The deterministic tolerances are transparent but cannot represent every perceptually acceptable alternative. Path comparison uses finite geometric samples and can miss narrow or highly localized differences; CIE76 is only an approximate color metric, and viewport-relative caps may be too strict or permissive for particular scenes. The semantic comparator does not execute arbitrary CSS, browser layout, animation, scripting, accessibility, or editor-specific behavior.

Instructions and acceptable edits were authored and checked by the benchmark authors; no independent annotator-agreement or human-performance study was run. The sensitivity analysis varies tolerance width but is not a substitute for human calibration. Consequently, binary reward should be interpreted as success under this released executable specification, not universal perceptual approval.

The evaluation uses one scored outcome per model–task pair and therefore cannot characterize sampling stability. OpenRouter can change prices, routing, model versions, and availability; requested and resolved IDs are published to make this visible. The “open-weight” grouping is manifest metadata, not a legal determination.

Twenty scenic source files in the frozen corpus lack recoverable per-file provenance metadata. We do not claim original authorship of those illustrations. They are retained to preserve the evaluated task set, marked as provenance pending in the artifact, and should be treated as research-only until provenance is resolved. This limits redistribution confidence and is a material artifact risk.

Finally, the benchmark evaluates text-and-source editing rather than multimodal screen interaction. Models see SVG code and a visual description, but not a raster preview or interactive editor. Performance therefore combines scene reasoning, source localization, and code generation under one proto-

col.

9 Conclusion

Vector-Bench isolates a practical requirement for editing agents: requested change is not enough; preservation is part of the specification. Author-written visual instructions make the task less like copying a patch, while executable SVG targets make evaluator decisions inspectable. Across 34 endpoints, partial repair is much more common than specification-faithful repair. The released outputs and diff reports make that gap inspectable at the object and source level and provide a compact target for improving structured visual editors.

References

- Josef Kuchař, Marek Kadlčík, Michal Spiegel, and Michal Štefánik. 2025. [VectorEdits: A dataset and benchmark for instruction-based editing of vector graphics](#). *Preprint*, arXiv:2506.15903.
- Kunato Nishina and Yusuke Matsui. 2024. [SVGEEdit-Bench: A benchmark dataset for quantitative assessment of LLM’s SVG editing capabilities](#). *Preprint*, arXiv:2404.13710.
- Kunato Nishina and Yusuke Matsui. 2025. [SVGEEdit-Bench V2: A benchmark for instruction-based SVG editing](#). *Preprint*, arXiv:2502.19453.
- OpenRouter. 2026. [Api reference and model catalog](https://openrouter.ai/docs/api-reference/overview). <https://openrouter.ai/docs/api-reference/overview>. Accessed 2026-07-20.
- Juan A. Rodriguez, Abhay Puri, Shubham Agarwal, Issam H. Laradji, Pau Rodriguez, Sai Rajeswar, David Vazquez, Christopher Pal, and Marco Pedersoli. 2023. [StarVector: Generating scalable vector graphics code from images and text](#). *Preprint*, arXiv:2312.11556.
- World Wide Web Consortium. 2018. [Scalable vector graphics \(SVG\) 2](#). Technical report, W3C.
- Ronghuan Wu, Wanchao Su, Kede Ma, and Jing Liao. 2023. [IconShop: Text-guided vector icon synthesis with autoregressive transformers](#). *Preprint*, arXiv:2304.14400.
- Ximing Xing, Juncheng Hu, Guotao Liang, Jing Zhang, Dong Xu, and Qian Yu. 2024. [Empowering LLMs to understand and generate complex vector graphics](#). *Preprint*, arXiv:2412.11102.

A Artifact Notes

Reproducibility. The release includes the frozen corpus, prompts, runner, evaluator, tests, analysis, paper source, Harbor tasks, and a chronological per-output trace viewer. Credentials and authorization headers are excluded. The released study includes sanitized final-response records and verifier snapshots; unavailable legacy retry and transport envelopes are explicitly labeled rather than inferred. New runs retain these events append-only. The public artifact is available at <https://www.vecbench.xyz>, with source at <https://github.com/yug-space/vector-edit-gym>.

Shared authorship. Yug Aditi Gupta and Prannay Hebbar contributed equally to the benchmark, artifact, analysis, and writing. The paper and accompanying material are shared work.

B Full Endpoint Results

Values are percentages. Errors include persistent provider failures and missing usable SVG output. Endpoint IDs, resolved IDs, response IDs, and per-task details are available in the machine-readable release.

Model	Group	Full	Near	Repair	Progress	Clean	UCR	Valid	Trunc.	Errors
Claude Sonnet 5	frontier	15.0	2.5	15.0	43.7	42.5	0.8	62.5	40.0	0.0
KAT Coder Air V2.5	cheap-control	12.5	0.0	15.0	27.3	25.0	0.3	42.5	57.5	0.0
MiniMax M3	open-weight	10.0	0.0	10.0	33.0	25.0	1.0	45.0	55.0	0.0
HY 3	open-weight	7.5	2.5	10.0	56.8	37.5	1.8	100.0	0.0	0.0
Gemma 4 31B IT	open-weight	5.0	5.0	10.0	54.8	47.5	0.8	87.5	5.0	0.0
Qwen3.5 397B A17B	open-weight	5.0	0.0	5.0	39.8	32.5	1.4	65.0	32.5	0.0
DeepSeek V4 Pro	open-weight	5.0	2.5	5.0	30.6	20.0	1.2	47.5	52.5	0.0
Kimi K2.6	open-weight	5.0	5.0	5.0	16.0	15.0	0.3	25.0	72.5	2.5
Qwen3 Coder	open-weight	2.5	0.0	5.0	54.7	45.0	1.4	100.0	0.0	0.0
Gemini 3.1 Flash Lite	cheap-control	2.5	0.0	7.5	54.5	25.0	2.5	100.0	0.0	0.0
Devstral 2512	open-weight	2.5	5.0	2.5	46.0	57.5	0.6	97.5	0.0	0.0
Inkling	open-weight	2.5	0.0	2.5	26.6	27.5	0.8	45.0	55.0	0.0
GPT-5	frontier	2.5	0.0	2.5	9.1	7.5	0.3	15.0	85.0	0.0
Nemotron 3 Ultra	open-weight	2.5	0.0	2.5	4.9	5.0	0.5	7.5	92.5	0.0
Claude Haiku 4.5	frontier	0.0	0.0	2.5	54.0	27.5	2.0	100.0	0.0	0.0
Seed 2.0 Mini	cheap-control	0.0	2.5	2.5	51.2	35.0	2.0	92.5	2.5	0.0
GPT-OSS 120B	open-weight	0.0	5.0	0.0	49.9	42.5	1.1	95.0	5.0	0.0
Llama 4 Maverick	open-weight	0.0	0.0	2.5	46.9	20.0	2.5	92.5	0.0	0.0
Mistral Small 2603	open-weight	0.0	0.0	0.0	39.0	5.0	4.7	95.0	0.0	0.0
GPT-5.4 Nano	cheap-control	0.0	0.0	0.0	38.0	22.5	2.2	97.5	0.0	0.0
GPT-OSS 20B	open-weight	0.0	2.5	2.5	36.5	27.5	1.6	72.5	27.5	0.0
Phi-4	open-weight	0.0	0.0	0.0	30.7	12.5	3.8	92.5	0.0	0.0
Laguna M.1	open-weight	0.0	2.5	0.0	23.3	25.0	1.4	42.5	55.0	0.0
Ling 2.6 Flash	cheap-control	0.0	0.0	0.0	19.4	15.0	4.8	87.5	5.0	0.0
Granite 4.1 8B	open-weight	0.0	0.0	0.0	18.5	10.0	14.3	77.5	2.5	0.0
GLM 4.7	open-weight	0.0	0.0	0.0	15.1	12.5	0.9	30.0	70.0	0.0
MiMo V2.5 Pro	open-weight	0.0	0.0	0.0	6.4	5.0	0.8	10.0	87.5	0.0
Qwen3.6 35B A3B	open-weight	0.0	0.0	2.5	4.9	2.5	1.4	7.5	92.5	0.0
Trinity Large Thinking	open-weight	0.0	0.0	0.0	1.8	0.0	0.0	2.5	97.5	0.0
Gemini 3.1 Pro Preview	frontier	0.0	0.0	0.0	0.0	0.0	–	0.0	100.0	0.0
Nemotron 3 Nano	open-weight	0.0	0.0	0.0	0.0	0.0	–	0.0	80.0	0.0
OLMo 3 32B Think	open-weight	0.0	0.0	0.0	0.0	0.0	–	0.0	0.0	100.0
Reka Flash 3	open-weight	0.0	0.0	0.0	0.0	0.0	–	0.0	12.5	0.0
Step 3.7 Flash	open-weight	0.0	0.0	0.0	0.0	0.0	–	0.0	100.0	0.0

C Representative Output Trace

Table 6 follows one published outcome through request, extraction, tolerant repair checks, preservation, validity, and binary reward. The example is useful because the generated handlebar is not the canonical path, yet its sampled geometry lies inside the released tolerance. Thus an acceptable approximation passes without weakening the requirement that unrequested content remain unchanged.

Trace event	Recorded value
Identity	Task sv_005; requested and resolved endpoint anthropic/claude-sonnet-5; trace legacy:anthropic/claude-sonnet-5:sv_005.
Instruction	“A stray price sticker has appeared in the market, one pear is red instead of green, and the bicycle handlebar has a distorted bend. Clean up those three issues while keeping the stalls and shoppers unchanged.”
Request	System repair protocol plus the corrupted SVG; provider-default temperature; 4,243-token output ceiling. The request used 3,700 prompt tokens.
Response	Response gen-1784563591-dtcYmLQV0qHEuDb7ucuZ; stop reason stop; 3,521 completion tokens; recorded cost \$0.04261.
Extraction	A complete SVG was extracted. This historical row is labeled legacy_final_record: the extracted SVG is retained, but the original response wrapper and retry envelopes were not retained and are not reconstructed.
Handlebar check	Expected path M122 96 C130 88 136 92 136 98; produced path M122 96 C128 90 132 90 134 96. Sampled-geometry distance $1.883 \leq 2.961$; check passed; repair progress 55.5%.
Pear check	Expected and produced fill #a3e635; $\Delta E_{76} = 0 \leq 18$; check passed; repair progress 100%.
Sticker check	Expected absence and produced absence; check passed; repair progress 100%.
Preservation	Every protected object passed semantic comparison; preservation 100%; valid-output UCR 0%.
Validity	The output parsed and passed standalone SVG validity checks.
Outcome	All three required repairs passed, semantic preservation passed, and validity passed. Binary reward = 1; mean soft repair progress = 85.2%.

Table 6: Chronological verifier trace for a published full pass. Soft progress retains information about approximate repair magnitude; it is not an additional binary gate once each requested edit is within tolerance.

The interactive release exposes the corresponding prompt, response artifact, per-edit distances, preservation report, and score history at <https://www.vecbench.xyz/traces/run> using task sv_005 and model anthropic/claude-sonnet-5.

D A Worked Failure Trace

The trace above shows a pass; this one shows why passes are rare. In Figure 8, the strongest endpoint in the study, Claude Sonnet 5, answers the station task this paper opens with (sv_001). The output looks plausible and is valid SVG. But only the signal is repaired ($\Delta E_{76} = 0$). The door overshoots the hidden target by 24 user units. The reshaped wire lands farther from the target than the corruption it was asked to fix. The model also drags the door handle, a protected object the corruption never touched, along with the door. One repair out of three plus one collateral change put the response in the “valid, incomplete repair” bucket that dominates Figure 5. The reward is zero. This is the paper’s central gap in a single trace: visible repair work, not the repair-and-preserve contract. The failure modes of Section 7 co-occur in one response. The released viewer exposes every per-check distance, tolerance, and rendered before/after.¹

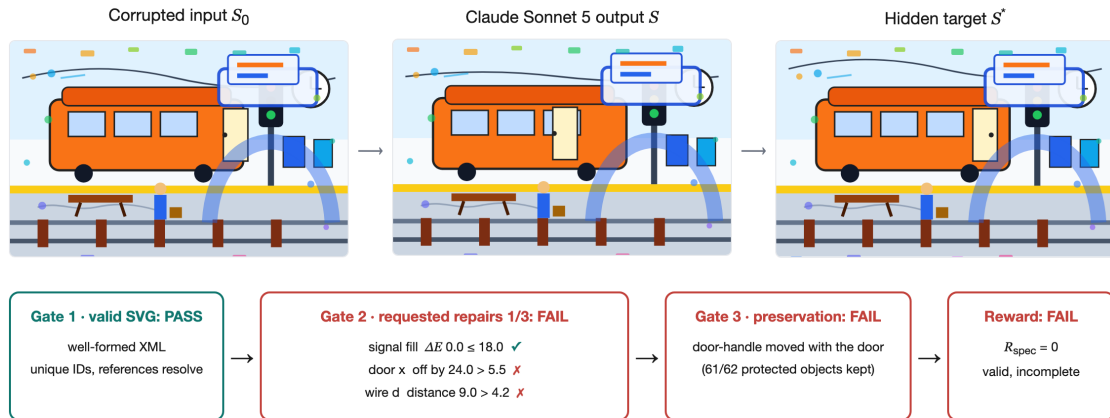


Figure 8: One scored trace, end to end. Top: corrupted input, model output, hidden target. Bottom: the three-gate evaluation. The document is valid (gate 1). Only the signal-fill repair meets its tolerance (gate 2). Moving the door handle changes a protected object (gate 3). The specification reward is zero.

¹https://www.vecbench.xyz/traces/run?task=sv_001&model=anthropic/claude-sonnet-5